DO NOT WRITE IN THIS AREA

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17 At a height of n metres above sea level, the atmospheric pressure is P_n millibars.
At a height of $(n + 1000)$ metres above sea level, the atmospheric pressure is $P_{(n + 1000)}$ millibars

where $P_{(n + 1000)} = 0.88 P_n$

At sea level, the atmospheric pressure is 1013 millibars.

Calculate the atmospheric pressure at a height of 3000 metres above sea level.
Give your answer correct to 3 significant figures.

1000 m: 1013 × 0.88 = 891.44

1 mark
One 0.88 multiplier for each 1000 m.

2000 m: 891.44 × 0.88 = 784.4672

3000 m: 784.4672 × 0.88 = 690.331136

Pressure = 690 millibars (3 s.f.)

1 mark
Repeat for all three increases.

1 mark
Round only at the end; 690 is given to 3 significant figures.

690 millibars

(Total for Question 17 is 3 marks)